Boostify ImageGen Platform - Project Requirements & Developer Handoff

Project Overview

We are building a comprehensive, user-friendly SaaS platform for AI-powered product image generation. The primary goal is to completely redesign and develop the frontend from scratch, integrating with our existing backend workflows (n8n webhooks) and Supabase database/authentication.

Timeline:

- **Start Date:** Immediate
- **Target Deadline:** March 16th, 2026 (1-week development sprint)

Core Architecture & Tech Stack

- **Frontend Framework:** Next.js (React) - App Router
- **Styling:** Modern, clean, professional UI (White/Light theme by default). Premium animations and glassmorphism where appropriate.
- **Authentication & Database:** Supabase
- **Workflow Automation:** n8n (External webhooks)
- **Object Storage:** AWS S3 (for storing generated images)

Key Features & Page Requirements

1. Authentication (Login / Signup)

- **Flow:** Users must sign up or log in to access the platform.
- **Provider:** Supabase Auth (Email & Password).
- **Default Credits:** New signups automatically receive $10 worth of free credits.
- **Route Protection:** Unauthenticated users should be redirected to the login page.

2. Pricing & Subscription Plans

The platform will offer the following tiers:

- **Free Plan:** $10 initial credits (Free)
- **Plus Plan:** High quality, $50 credits/month — **₹250 INR**
- **Pro Plan:** High quality, $100 credits/month — **₹499 INR**
- **Enterprise Plan:** Custom workflows, very high quality — **Contact us at contact@boostifycorp.dev**

3. Settings Page (Credentials Management)

Users need a secure place to store their integration credentials. These must be saved to the Supabase database (`imagegen_user_settings` table) and linked to their `user_id` .

- **AWS S3 Configuration:** Access Key ID, Secret Access Key, Region, Bucket Name.
- **n8n Webhook URLs:** Input fields for the Amazon PDP Webhook URL and the Single Image Webhook URL.
- **Cloud Sync:** Settings should auto-fill from Supabase when the user logs in on any device.

4. Image Editor (Single Image Generation)

- **Uploads:** Allow up to 3 reference images (minimum 1 required).
- **Settings:** Prompt input, negative prompt, aspect ratio, style presets.
- **Execution:** Sends a payload containing the images, prompt, logic parameters, and webhook URL to the user-defined n8n webhook.
- **UI UX:** Display a loading state with an animated progress bar and elapsed time while waiting for the webhook response.

5. Amazon PDP (Catalogue Generation)

- **Input:** Amazon Product URL/ASIN.
- **Execution:** Triggers the Amazon PDP webhook in n8n.
- **UI/UX:** Similar premium loading states and parameter configurations as the Image Editor.

6. Gallery

- **Functionality:** View previously generated images.
- **Storage:** Fetched directly from the user's configured AWS S3 bucket.
- **Features:** Lightbox view, download buttons, pagination/infinite scroll.

Design Guidelines

- **Theme:** White/light theme by default. Clean, minimal, spacious, and professional.
- **Typography:** Modern sans-serif fonts (Inter, SF Pro, Geist).
- **Animations:** Include smooth, purposeful animations (hover states, loading spinners, page transitions).
- **Responsiveness:** Must work flawlessly on all screen sizes (Mobile, Tablet, Desktop).

Developer Instructions

1. **GitHub Setup:** Clone the repository. The collaboration invite has been sent to Boostify.
2. **Environment Variables:** Set up your `.env.local` with Supabase URLs/Keys.
3. **Database Setup:** Ensure the Supabase migration script (`supabase-migration.sql`) has been executed to set up the `imagegen_user_settings` table and Row Level Security (RLS) policies.
4. **Code Quality:** Ensure modular, reusable components. Document complex logic.

Contact: If you face any blockers, reach out immediately to ensure we hit the March 16th deadline.